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Roll no.

B.Tech, Electrical Engineering Semester: 2nd
END-SEMESTER EXAMINATION (SUMMER), July 2026

Course Title: **SIGNAL AND SYSTEMS**

Course Code: EEEEC03

Duration: 3:00 Hours

Max. Marks: 50

Note: - Attempt all five questions. Missing data/information, if any, may be suitably assumed and mentioned in the answer.

Q.No	Question	Marks	CO
Q. 1	Attempt any 2 parts of the following		
(a)	Determine whether the given signal is an energy signal or a power signal: $x(t) = e^{-2t}u(t)$	5	CO-1
(b)	(i) Explain time-domain representation of an LTI system. (ii) Explain signal scaling and shifting.	5	CO-1
(c)	Evaluate: $y(t) = x(t) * h(t)$ Where, $x(t) = u(t), h(t) = e^{-t}u(t)$	5	CO-1
Q. 2	Attempt any 2 parts of the following		
(a)	Find $L^{-1}\left\{\frac{s+3}{(s+1)(s+2)}\right\}$	5	CO-2
(b)	Find the poles and zeros of $H(s) = \frac{s+2}{(s+1)(s+4)}$	5	CO-2
(c)	Find the impulse response of the system represented by $H(s) = \frac{2s+5}{s^2+5s+6}$	5	CO-2
Q. 3	Attempt any 2 parts of the following		
(a)	A series RL circuit has $R = 5\Omega, L = 2H$. A DC voltage of 10 V is applied at $t = 0$. Determine the current $i(t)$.	5	CO-3

(b)	(i) Find the poles and zeros of $H(s) = \frac{s+2}{s^2+5s+6}$ (ii) Determine the driving-point impedance of a network if $V(s) = 10(s+2), I(s) = 2(s+5)$	5	CO-3
(c)	A series RLC circuit has $R = 5\Omega, L = 1H, C = 0.25F$. Determine the differential equation and solve for the current.	5	CO-3
Q. 4	Attempt any 2 parts of the following		
(a)	Define Z-Transform. State the properties of Z-Transform.	5	CO-4
(b)	Determine the frequency response of $H(z) = \frac{1-z^{-1}}{1-0.8z^{-1}}$	5	CO-4
(c)	i) Find: $Z\{a^n\}$ ii) Find: $Z\{nu[n]\}$	5	CO-4
Q. 5	Attempt any 2 parts of the following		
(a)	Find the Fourier Series of the half-wave rectified waveform shown below. The waveform is defined as $x(t) = \begin{cases} V_m \sin \omega t, & 0 < \omega t < \pi \\ 0, & \pi < \omega t < 2\pi \end{cases}$ with period $T = \frac{2\pi}{\omega}$	5	CO-5
(b)	Find the amplitude spectrum and phase spectrum of $x(t) = 4\cos(50t) + 3\sin(100t)$	5	CO-5
(c)	i) Explain trigonometric Fourier Series. ii) Explain exponential Fourier Series.	5	CO-5